MIL-P-60125A(MU)
10 January 1968
SUPERSEDING
MIL-P-60125(MU)
24 August 1964

MILITARY SPECIFICATION

PROJECTOR, RETICLE IMAGE: M14

1. SCOPE

1.1 This specification covers one type of electro-mechanical unit known as Reticle Image Projector: M14, hereinafter referenced as projector (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F10516310	Projector, Reticle Image: M14

FSC 1240

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PACKAGING DATA SHEET

MIL-P-14232/P10516310 Packaging of Projector, Reticle Image: M14

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The projector shall be manufactured in accordance with Drawing F10516310, "Projector, Reticle Image, M14" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical, optical and electrical characteristics, performance, and permanence to insure inherent quality and performance capabilities

3.3 General specification. The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- The projector shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of + 60° to 90°F after having been exposed to and thermally stabilized at ambient temperature of -80°F and +160°F.

3.4.2 Extreme operating temperature.- The projector shall meet all the applicable performance requirements of 3.7 and 3.8 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4.3 Vibration.- The projector shall meet the requirements of 3.4.4 to 3.10 inclusive after being vibrated in a vertical plane for five minutes plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second.

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3.4.4 Sealing.- The projector shall show no evidence of leakage in excess of 0.2 pounds per square inch gage (psig) when subjected to an internal pressure of 5.0 ± 0.10 psig for a minimum of 2 hours.

3.4.5 Cleanliness.- Subsequent to vibration, the projector shall meet the cleanliness requirements of MIL-O-13830. There shall be no more than one particle of dirt visible on the reticle. The area of the particle shall not be in excess of the area equivalent to a size 10 dig.

3.5 Orientation.- The projector shall be adjusted to the following conditions and energized in accordance with Figure I of this specification.

(a) The bottom surface of Key 10515975 (in Housing 10516262) shall be positioned horizontal within plus or minus 15 minutes.

(b) The mounting surfaces above and below the key shall be positioned in a vertical plane within one degree.

(c) With electrical zero (null setting .0000 on ratio standard) established, the line of sight through the center of the reticle projector boresight circle shall be adjusted by the boresight knobs to be perpendicular to the mounting surfaces within 0.15 mils and the scales slipped to index "(4)".

3.6 Infinity sight.

3.6.1 Infinity sight reticle.- With the projector positioned as in 3.5 the line of sight of the center of the infinity sight reticle circle shall be adjusted to subtend an angle of 26 degrees plus or minus 15 minutes with the horizontal plane of the locating key and shall lie in a vertical plane perpendicular to the vertical mounting surface as adjusted by the boresight knobs.

3.6.2 Reticle focus.- The focus of the infinity sight reticle shall lie between -0.75 and -1.0 diopter as measured with a dioptometer.

3.6.3 Reticle accuracy.- The inner diameter of the reticle circle shall subtend an angle of 21 mils plus or minus three mils.

3.6.4 Boresight knob travel.- Rotation of the boresight knobs for the infinity sight shall cause the reticle to move 10 mils to the right and 10 mils to the left, and 10 mils above and 10 mils below the position established in 3.6.1.

3.7 Reticle projector.

3.7.1 Continuity.- The continuity of the projector shall conform to the schematic Drawing D10516185.

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3.7.2 Reticle tilt.- The reticle line shall be parallel to a horizontal target line within 1.0 degree.

3.7.3 Reticle accuracy.- The total angular subtense at the extremities of the series of horizontal reticle lines through the boresight circle shall be 29.5 mils plus or minus 1.0 mil.

3.7.4 Parallax.- Parallax between the image of the ballistics reticle in the reticle projector and the image in an auxiliary telescope adjusted for zero parallax at infinity shall not exceed 0.1 mil.

3.7.5 Reticle movement.- Power operation shall move the reticle through a total horizontal excursion of 86.4 ± 0.7 mils.

3.7.6 Boresight knob travel.- With the projector positioned as in 3.5, rotation of the elevation boresight knob shall move the reticle a minimum of 11.7 mils in elevation and 17.6 mils in depression. Rotation of the deflection boresight knob shall move the reticle a minimum of 14.6 mils to the right and 14.6 mils to the left.

3.7.7 Reticle shift.- Any movement of the elevation and deflection boresight knobs shall not cause the reticle to shift by more than 0.3 mil in a direction perpendicular to the intended movement of the reticle cell including backlash.

3.7.8 Backlash.- Backlash in the elevation and deflection boresight knobs shall not exceed 0.15 mil.

3.7.9 Motor - Tachometer fixed phase excitation.- Apply 115 volts ± 1%, 400 CPS ± 1% between pins A and B of J1 for excitation of the fixed phase of the motor-tachometer.

3.7.9.1 Motor control phase excitation.- When energized as in 3.7.9, the application of 48 volts ± .5 volt 400 CPS ± 1% between pins C and F of J1, shall move the reticle to the right when the voltage at pin F is in phase with the voltage at pin A. When the voltage between pins A and F is 180 degrees out of phase, the reticle shall move to the left.

3.7.9.2 Slew speed.- When energized as specified in 3.7.9.1, the reticle shall slew from stop to stop in both directions within a maximum running time of 2.0 seconds for each direction at standard ambient temperature and within 5.0 seconds at the extreme operational temperatures.

3.7.9.3 Tachometer output voltage.- Under the slew speed conditions of 3.7.9.2, the tachometer output voltage at pin H of J1 shall be a minimum of 7.5 volts peak to peak and in phase with the voltage at pin A when the reticle is moving toward the right with projector mounted as in 3.5.

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3.7.10 Electro-optical accuracy.- This requirement shall be met with the projector positioned in accordance with 3.5 and energized in accordance with Figure I of this specification. The mean value of at least 3 sets of bi-directional horizontal reticle line of sight readings each side of electrical 0 shall be 36.5 ± .4 mils at the null settings of .83052. All readings at null settings .83052 on each side of electrical 0 shall be within 0.15 mil of their mean values.

3.7.10.1 Reticle traversing accuracy.- To determine linear traversing accuracy at null settings listed in Table I, at least 3 sets of bi-directional horizontal reticle line of sight readings each side of electrical 0 shall be taken. All reticle line of sight readings for each null setting shall be within 0.15 mil of the applicable reticle position values computed per Table I.

TABLE I

Null Settings	Reticle Position (Mil)
.83052	*R = X; L = Y
.66442	R = 80%X; L = 80%Y
.33221	R = 40%X; L = 40%Y
.16610	R = 20%X; L = 20%Y

*R and L = right and left of electrical. X and Y = mean reticle line of sight value established in paragraph 3.7.10.

3.7.10.2 Zero line of sight repeatability.- The reticle line of sight at electrical 0 shall remain perpendicular with mounting surfaces within 0.15 mil for at least 3 sets of bi-directional readings.

3.7.11 Illumination.- A 24 volt DC power source, when applied to pins G and Z (ground) of connector J-1, shall illuminate the infinity sight reticle patterns when viewed in a darkened area, the pattern shall appear uniform and free of ragged edges or breaks throughout their entire area. Counterclockwise rotation of the rheostat shall vary the intensity of the reticle image in the reticle projector and the infinity sight from high to low intensity.

3.8 Operability.

3.8.1 Infinity sight.- The maximum torque required to rotate the azimuth and elevation boresightknobs for the infinity sight shall not exceed five pound inches at standard ambient temperatures and 10 pound inches at the extreme temperatures specified in 3.4.2.

3.8.2 Reticle projector.

3.8.2.1 Disengagement force.- The force required to disengage the deflection and elevation boresight knobs from their normal position shall not be less than one pound nor more than three pounds at standard ambient temperature.

3.8.2.2 Boresight knob torque.- The torque required to rotate the

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boresight knob shall not exceed four pound-inches nor be less than one half pound-inches at standard ambient temperature. This value shall not change by more than one and one half (1-1/2) pound inch at the temperatures of 3.4.2.

3.8.2.3 Rheostat torque.- With the projector at standard ambient temperatures, the torque required to rotate the rheostat knob shall not exceed four pound inches. The value shall not change by more than 12 pound inches at the temperatures of 3.4.2.

3.9 Insulation.

3.9.1 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the projector are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than five seconds.

3.9.2 Insulation resistance.- All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 meghoms with 500 volts direct current (vdc) applied.

3.10 Interior atmosphere.- The interior atmosphere of the accepted projector shall contain nitrogen having a dew point at least as low as -25°F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three units. The sample shall be manufactured in the same manner, using the same materials,

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equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained is Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III, and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III, and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% defective		
101. Vibration	3.4.3	4.6.3
102. Cleanliness	3.4.5	4.6.4
103. Sealing	3.4.4	4.6.5
104. Interior atmosphere	3.10	4.6.6
105. Projector orientation	3.5	4.6.7, 4.6.7.1, 4.6.7.2
106. Infinity sight reticle	3.6.1	4.6.7.1.1
107. Reticle focus	3.6.2	4.6.7.1.4
108. Reticle accuracy, infinity sight	3.6.3	4.6.7.1.2
109. Boresight knob travel, infinity sight	3.6.4	4.6.7.1.3
110. Reticle tilt	3.7.2	4.6.7.3.1
111. Reticle accuracy, reticle projector	3.7.3	4.6.7.2.2

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TABLE II - CLASSIFICATION OF DEFECTS (CONTINUED)

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.0% defective		
112.	Parallax	3.7.4	4.6.7.2.3
113.	Boresight knob travel, reticle projector	3.7.6	4.6.7.2.4
114.	Backlash	3.7.8	4.6.7.2.5
115.	Reticle shift	3.7.9.1	4.6.7.2.6
116.	Reticle movement	3.7.5	4.6.7.2.7
117.	Motor control phase excitation	3.7.9.1	4.6.7.2.8
118.	Slew speed	3.7.9.2	4.6.7.2.9
119.	Tachometer output voltage	3.7.9.3	4.6.7.2.10
120.	Electro-optical accuracy	3.7.10	4.6.7.2.11
121.	Reticle traversing accuracy	3.7.10.1	4.6.7.2.12
122.	Zero line of sight repeatibility	3.7.10.2	4.6.7.2.13
123.	Illumination	3.7.11	4.6.7.2.14
124.	Boresight knob torque, infinity sight	3.8.1	4.6.8
125.	Boresight knob disengagement, reticle projector	3.8.2.1	4.6.9
126.	Boresight knob torque, reticle projector	3.8.2.2	4.6.10
127.	Rheostat knob torque	3.8.2.3	4.6.11
128.	Continuity	3.7.1	4.6.12
129.	Dielectric strength	3.9.1	4.6.13
130.	Insulation resistance	3.9.2	4.6.14
MINOR:	NONE		

NOTE: The tests of Table II shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One reticle projector shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawing Visual
302.	General specification	3.3	MIL-F-13926 - Visual

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4.4.2 Environmental.- Three reticle projectors shall be selected at random as samples from each 50 produced, or from each months production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV, and shall then meet the requirements and tests in Table II.

TABLE IV

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1
304.	Reticle tilt	3.4.2, 3.7.2	4.6.2, 4.6.7.2.1
305.	Reticle accuracy, reticle	3.4.2, 3.7.3	4.6.2, 4.6.7.2.2
306.	Parallax	3.4.2, 3.7.4	4.6.2, 4.6.7.2.3
307.	Boresight knob travel, reticle projector	3.4.2, 3.7.6	4.6.2, 4.6.7.2.4
308.	Backlash	3.4.2, 3.7.8	4.6.2, 4.6.7.2.5
309.	Reticle shift	3.4.2, 3.7.7	4.6.2, 4.6.7.2.6
310.	Reticle movement	3.4.2, 3.7.5	4.6.2, 4.6.7.2.7
311.	Motor control phase excitation	3.4.2, 3.7.9.1	4.6.2, 4.6.7.2.8
312.	Slew speed	3.4.2, 3.7.9.2	4.6.2, 4.6.7.2.9
313.	Tachometer output voltage	3.4.2, 3.7.9.3	4.6.2, 4.6.7.2.10
314.	Electro-optical accuracy	3.4.2, 3.7.10	4.6.2, 4.6.7.2.11
315.	Reticle traversing accuracy	3.4.2, 3.7.10.1	4.6.2, 4.6.7.2.12
316.	Zero line of sight repeatability	3.4.2, 3.7.10.2	4.6.2, 4.6.7.2.13
317.	Illumination	3.4.2, 3.7.11	4.6.2, 4.6.7.2.14
318.	Boresight knob torque, infinity sight	3.4.2, 3.8.1	4.6.2, 4.6.8
319.	Boresight knob disengagement, reticle projector	3.4.2, 3.8.2.1	4.6.2, 4.6.9
320.	Boresight knob torque, reticle projector	3.4.2, 3.8.2.2	4.6.2, 4.6.10
321.	Rhoestat knob torque	3.4,2, 3.8.2.3	4.6.2, 4.6.11
322.	Continuity	3.4.2, 3.7.1	4.6.2, 4.6.12

NOTE: The tests for Characteristics 304 through 322 inclusive shall be conducted at -40° and +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- The pressure testing equipment to test sealing of the projector shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Orientation of projector (mounting).- The testing equipment shall position the vertical mounting surfaces of the reticle projector in a vertical plane within one degree, and the bottom surface of the Key 10515975 in a horizontal plane within plus or minus 15 minutes. The testing equipment shall provide for positioning an observation telescope (see 4.5.2.3.3) at the exit window (Window 10515965) of the reticle projector. The optical axis of the telescope shall be established perpendicular to the vertical mounting surface of the projector within 30 seconds of arc. The testing equipment shall also provide for positioning the observation telescope at the exit window (Window 8229032) of the infinity sight. The optical axis of the telescope shall be inclined to form an angle of 26 degrees plus or minus 15 minutes with the horizontal plane that contains the bottom surface of Key 10515975, and lie in a vertical plane within 30 seconds to the vertical mounting surface of the projector.

4.5.2.3.3 Reticle positioning.- Electronic circuitry conforming to Figure 1 that also incorporates the test equipment specified in 4.5.3 shall be utilized to energize the reticle projector, and position its reticle assembly as required.

4.5.2.3.4 Optical accuracies.- The observation telescope, utilized to measure the optical accuracies of the projector, shall be at least 10 power (10X) with at least a 2 inch entrance pupil, and have a field of view of at least 90 mils. The reticle of the telescope shall contain a horizontal and vertical line with graduation marks at 0.5 mil increments on the reticle lines up to 45 mils on each side of the vertical and horizontal lines. The telescope shall be positioned at infinity focus. The telescope shall be

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adjusted for zero parallax at the infinity setting. The telescope shall be capable of measuring the appropriate optical accuracies of the projector at -40°F, +60° to +90°F, and +150°F.

4.5.2.3.5 Force requirements (disengagement).- Testing equipment for measuring direct force (pull) with a minimum range of zero to five pounds, the accuracy of which shall be plus or minus two ounces throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with a suitable adapter to accommodate testing of the boresight knobs.

4.5.2.3.6 Torque requirements.- A torque wrench type tester with a minimum range from zero to fifteen inch pounds, the accuracy of which shall be plus or minus one-quarter inch pound throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with suitable adapters to accommodate testing of the boresight knobs and rehostat knob.

4.5.2.3.7 Set-up and calibration.- Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following stamdard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

Item Number	Description
1. A. C. Ratio Standard	Ratio range from .00000 to 1.00000 accuracy to ± .01%. Input impedance 500,000 ohms minimum at 400 cps. Series impedance 10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance - 1 megohm or greater
3. A. C. Power Supply	0-130 volts rms 400 cps ± 1%. Harmonic distortion 2% maximum. Output current 1 ampere minimum.
4. D. C. Power Supply	0.30 volts ± 5% maximum. Ripple 0.1 volt peak to peak max. Output current 2 amperes minimum.

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4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The projector shall be exposed to each temperature extreme specified in 3.4.1 for a minimum of 5 hours. After each temperature exposure the projector shall be stabilized in an ambient temperature between +60° and +90°F and subjected to the visual, tactile, and audible examination of 4.6.3.1. The projector shall then be subjected to the tests in 4.6.2 through 4.6.14 inclusive to determine conformance to 3.4.1. Caution shall be exercised during the temperature cycling to avoid subjecting the projector to thermal shock.

4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The projector shall be exposed to each temperature extreme specified in 3.4.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the projector shall be subjected to the applicable test of 4.6.7.2 through 4.6.11 inclusive. The projector shall conform to the requirements of 3.4.2. Caution shall be exercised during this test to avoid subjecting the projector to thermal shock. The test circuitry utilized to energize the projector shall be positioned external to the environmental test chamber.

4.6.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.3 shall be utilized for the vibration test. Unless otherwise specified the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. The projector shall be mounted on the vibrating table by means of an adapter. This adapter shall position the projector with its mounting surface in a vertical plane. The projector shall be vibrated in a vertical plane for a period of 5 minutes ± 15 seconds. Upon completion of the vibration cycle the projector shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration the projector shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as boresight knobs, reheostat knob, etc. An audible examination shall be made of the projector by shaking it and listening for loose internal parts or broken optical components, such as lenses, prism, reticles, etc. Any one of the possible defects defined above shall be cause to consider a projector defective. Projectors which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.14 inclusive to determine overall compliance with 3.4.3.

4.6.4 Cleanliness.- Energize the reticle projector and sight infinity lamp circuits in accordance with Drawing D10516185 for this test. The optical

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surfaces shall be inspected for cleanliness in accordance with MIL-O-13830. The projector shall conform to the cleanliness requirements of 3.4.5. Dirt particles which interfere with the legibility or accuracy of the reticle patterns shall be cause for rejection.

4.6.5 Sealing.- The internal pressurizing of the projector to test sealing, shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied through the valve in the projector. Pressurize the projector with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 2 hours. Within the first 60 minutes of this 2 hour time period, each elevation and deflection boresight knob shall be rotated through a minimum of 3 full excursions (cw stop-to ccw-to cw stop) of their rotational range. Any evidence of leakage in the projector at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.4.4, as measured on the pressure gauges of the testing equipment.

4.6.6 Internal atmosphere.- Upon satisfactory completion of the sealing test the projector shall be bled to 1 psig and flushed with dry nitrogen gas through a regulator that will prevent the internal pressure from exceeding the 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the projector has a dew point at least as low as minus 25 degrees Fahrenheit, as measured with a dew point tester. The projector shall then be sealed. The internal atmosphere of the projector shall conform to the requirements of 3.10.

4.6.7 Orientation.- The following tests shall be performed with the projector positioned on special testing equipment conforming to 4.5.2.3.2. The projector shall be energized in accordance with 4.5.2.3.3. Illuminate the projector reticle patterns.

4.6.7.1 Infinity sight.- Position the observation telescope (see 4.5.2.3.3) at the infinity sight exit window of the projector. Adjust the illumination of the infinity sight reticle pattern for best intensity.

4.6.7.1.1 Infinity sight reticle.- Superimpose the image of the infinity sight reticle, by means of its elevation and deflection boresight knobs, on the reticle of the observation telescope so that its vertical and horizontal reticle lines bisect the infinity sight circle. The infinity sight reticle shall meet the adjustment requirements of 3.6.1.

4.6.7.1.2 Reticle accuracy.- Perform this test with the infinity sight reticle positioned as in 4.6.7.1.1. The inner diameter of the infinity sight reticle shall conform to the requirements of 3.6.3, as measured on the reticle of the observation telescope.

4.6.7.1.3 Boresight knob travel (infinity sight).- Perform this test with the infinity sight reticle initially positioned as in 4.6.7.1.1. While

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observing the infinity sight circle rotate the azimuth boresight knob to the limit of rotation and note the amount of reticle movement, as measured on the observation telescope reticle. Return the infinity sight reticle to its initial starting position, then rotate the azimuth boresight knob in the opposite direction to its limit of rotation and note the amount of reticle movement. Return the infinity sight reticle to its initial starting position. Utilizing the above procedure move the reticle to its limit of travel in each direction (up and down) by means of the elevation boresight knob and note the amount of reticle movement. The amount of reticle movement (left, right, up and down) from the position established in 4.6.7.1.1 shall conform to the requirements of 3.6.4, as measured on the reticle of the observation telescope.

4.6.7.1.4 Reticle focus.- Remove the observation telescope from the infinity sight exit window. The reticle focus test shall be made with the aid of a calibrated dioptometer that has a magnification of at least 3 power (3X), and a diopter measuring range of at least plus and minus 2 diopters. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle, and set the dioptometer diopter scale to zero. Position the dioptometer of the exit window of the infinity sight. Obtain best focus of the infinity sight reticle by adjustment of the dioptometer diopter scale. The infinity sight reticle shall be in best focus within the limits specified in 3.6.2, as measured on the dioptometer diopter scale.

4.6.7.2 Reticle projector.- Position the observation telescope at the reticle projector exit window (Window 10515965). Adjust the illumination of the projector's ballistic reticle pattern for best intensity. Set a null setting of .00000 on the ratio standard and rotate potentiometer R7 of the test circuit until a null is produced on the oscilloscope. By means of the projector's elevation and deflection boresight knobs obtain coincidence of the intersecting point of the vertical and horizontal reticle lines of the ballistic reticle and observation telescope reticle. With the reticles in coincidence slip the boresight knob scales to index "(4)". Any additional adjustment (slipping) of the boresight scales during the tests of the projector is not permitted.

NOTE: For purposes of these tests the series of horizontal dash lines or spaces that bisect the circle on the ballistic reticle will be referred to as the horizontal reticle line.

4.6.7.2.1 Reticle tilt.- Utilizing the elevation boresight knob obtain coincidence of one extremity of the projector's horizontal reticle line with that of the horizontal reticle line in the telescope. Observe for coincidence at the opposite end of the horizontal line. For this test the angular subtense for one degree is equal to 3 thicknesses of the ballistic reticle line, as observed through the telescope. The ballistic horizontal reticle line shall be parallel to the horizontal reticle of the telescope within the tolerance specified in 3.7.2, as measured on the ballistic reticle.

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4.6.7.2.2 Reticle accuracy.- Perform this test in sequence to the test in 4.6.7.2.1. The measured distance between the extremities of the horizontal reticle line of the ballistic reticle shall equal the mil value specified in 3.7.3, as measured on the reticle of the observation telescope.

4.6.7.2.3 Parallax.- Parallax shall be measured at the center of the field of the ballistic reticle. Position the dot on the ballistic reticle adjacent to the vertical and horizontal reticle lines of the telescope. Any apparent movement of the ballistic reticle dot with respect to the telescope reticle lines when the observers head is moved side to side, then up and down, shall not exceed the tolerance specified in 3.7.4, as measured on the reticle of the observation telescope.

4.6.7.2.4 Boresight knob travel.- Position the reticles as in 4.6.7.2. While observing the ballistic reticle dot with reference to observation telescope reticle, rotate the elevation and deflection boresight knobs, in turn, to their clockwise stops then to their counterclockwise stops. At each of the knob stop positions the position of the ballistic reticle dot shall be noted with reference to the mil graduations on the telescope reticle. Reticle movement resulting from this rotation of the elevation and deflection boresight knobs shall conform to the requirements of 3.7.6, as measured on the reticle of the observation telescope.

4.6.7.2.5 Backlash.- Rotate the deflection boresight knob in a continuous clockwise direction, with backlash excluded, until the ballistic reticle dot is in coincidence with the vertical reticle line of the observation telescope. Record the position of the knob with reference to its index. Proceed to rotate the knob in the same clockwise direction for at least one-half turn, then reverse rotation of the knob (with backlash included) until the ballistic reticle dot is again in coincidence with the vertical reticle line of the telescope, without overtravel. Read the mil scale of knob. Any difference between the readings as measured on the knob scale is the amount of backlash. Repeat this operation with the elevation boresight knob utilizing the ballistic reticle dot and horizontal reticle line of the telescope. Backlash in the elevation and deflection boresight mechanism shall not exceed the tolerance specified in 3.7.8.

NOTE: Should overtravel occur during these tests for backlash the test shall be nullified and the procedure specified for backlash shall again be performed.

4.6.7.2.6 Reticle shift.- Conduct this test in conjunction with the test in 4.6.7.2.5. Should the ballistic reticle dot shift in deflection during the rotation of the elevation boresight mechanism, the amount of deflection shift shall be measured on the deflection boresight knob scale. Elevation shift of the reticle dot during rotation of the deflection boresight mechanism shall be measured on the elevation boresight knob scale. Any reticle shift resulting from the clockwise and counterclockwise rotation of the elevation and deflection boresight mechanisms shall not exceed the tolerance specified in 3.7.7, as measured on the applicable boresight knob.

NOTE: Upon completion of the tests of the boresight mechanisms, the ballistic reticle, boresight knobs, and knob scales of the projector shall be returned to applicable positions established in 4.6.7.2.

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4.6.7.2.7 Reticle movement.- Remove the test circuitry connections of Figure 1 from pins C and F of J1 for this test. Energize the motor tachometer as specified in 3.7.9. Apply the voltage specified in 3.7.9.1 between pins C and F of J1, with the voltage at pin F in phase with the voltage at pin A. Note the position of the ballistic reticle dot with reference to the mil graduations on the telescope reticle. Change the applied voltage at pins C and F so that the voltage at pin F is 180 degrees out of phase with the voltage at pin A. Again note the position of the ballistic reticle dot with reference to the mil graduations on the telescope reticle. The total horizontal excursion of the ballistic reticle dot shall meet the requirements of 3.7.5, as measured on the observation telescope reticle. The phase relationship of the voltages between pin A and pin F shall be determined by means of an oscilloscope.

4.6.7.2.8 Motor control phase excitation.- Conduct this test in conjunction with the test in 4.6.7.2.7. Reticle movement shall conform to the requirements of 3.7.9.1 when the phase relationship of the voltages between pin A and pin F is as specified.

4.6.7.2.9 Slew speed.- Conduct this test in sequence to the test in 4.6.7.2.7 with the aid of a standard calibrated time measuring instrument that conforms to the accuracy requirements of 4.5.2.2. Apply the voltage specified in 3.7.9.1 between pins C and F of J1, with the voltage at pin F in phase with the voltage at pin A. While observing the ballistic reticle dot change the applied voltage at pins C and F so that the voltage at pin F is 180 degrees out of phase with the voltage at pin A. The maximum running time required for the ballistic reticle to traverse from the right stop to the left stop shall not exceed the time period specified in 3.7.9.2, for the temperature indicated. Again while observing the ballistic reticle, change the voltage at pins C and F so that the voltage at pin F is in phase with the voltage at pin A. The maximum running time required for the ballistic reticle to traverse from the left stop to the right stop shall not exceed the time period specified for the temperature indicated.

4.6.7.2.10 Tachometer output voltage.- Conduct this test in sequence to the test in 4.6.7.2.9, utilizing an oscilloscope for measuring the output voltage of pin H of J1. Apply the voltage specified in 3.7.9.1 between pins C and F of J1, with the voltage at pin F 180 degrees out of phase with the voltage of pin A. Verify the position of the ballistic reticle at its left stop. Change the voltage at pins C and F so that the voltage of pin F is in phase with the voltage at pin A, and measure the output voltage at pin H of J1. Verify the position of the ballistic reticle at its right stop. The measured output voltage at pin H, and its phase relationship with respect to the voltage at pin A shall conform to the requirements of 3.7.9.3. The oscilloscope utilized in this test shall conform to the accuracy requirements of 4.5.2.2. Upon completion of the tests of the motor tachometer remove the voltage specified in 3.7.9.1 from pins C and F of J1.

4.6.7.2.11 Electro-optical accuracy.- Reconnect the test circuitry connections of Figure 1. Set a null setting of .00000 on the ratio standard and rotate potentiometer R7 of the test circuit until a null is produced on the oscilloscope. Position the test circuit phase switch for movement

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of the ballistic reticle to the right of electrical zero. Set a null setting of .83052 on the ratio standard and rotate potentiometer R7 in one direction (plus) without overtravel until a null is produced on the oscilloscope. Measure the deflection setting of the ballistic reticle dot with reference to the mil scale of the telescope reticle, and record this measurement as a plus mil value (R+) for purposes of calculating the mean value for the null setting of .83052. Set a null setting of .99999 on the ratio standard and rotate R7 in the same direction (plus) until a null is produced on the oscilloscope, then set .83052 on the ratio standard and rotate R7 in the opposite direction (minus) until a null is produced on the oscilloscope. Measure the deflection setting of the ballistic reticle dot and record this measurement as a minus mil value (R-). Set a null setting of .00000 on the ratio standard and rotate R7 in the same direction (minus) until a null is produced on the oscilloscope. Set .83052 on the ratio standard and using the above procedure obtain 2 additional bi-directional (plus and minus) mil deflection settings for the null setting of .83052. The mean mil value of the 3 sets of bi-directional readings for a null setting of .83052 shall be obtained by the arithmatical summation of all the R+ and R- divided by 6.

$$\text{i.e.: } \frac{\Sigma(R+) + \Sigma(R-)}{6} = \text{Right mean value}$$

The mean mil value for a null setting of .83052 on the right side of electrical zero, and each of the 6 measured deflection settings shall conform to the requirements of 3.7.10. Set a null setting of .00000 on the ratio standard and rotate potentiometer R7 until a null is produced on the oscilloscope. Position the test circuit phase switch for movement of the ballistic reticle to the left of electrical zero. Set a null setting of .83052 on the ratio standard, and utilizing the above procedure obtain the mean value for 3 sets of bi-directional (plus and minus) mil deflection settings for the null setting of .83052. The mean mil value for a null setting of .83052 on the left side of electrical zero, and each measured of the 6 measured deflection settings shall conform to the requirements of 3.7.10.

4.6.7.2.12 Reticle traversing accuracy.- Conduct this test in sequence to the test in 4.6.7.2.11. Utilizing the right mean mil value (X) and the left mean mil value (Y) obtained in 4.6.7.2.11, calculate and record the right and left reticle position (mil) for each of the null settings tabulated in Table 1 of 3.7.10.1. Position the test circuit phase switch for movement of the ballistic reticle to the right of electrical zero. Set a null setting of .00000 on the ratio standard and rotate the test circuit potentiometer R7 until a null is produced on the oscilloscope. Set a null setting of .16610 on ratio standard and rotate potentiometer R7 in one direction (plus) without overtravel until a null is produced on the oscilloscope. Measure the deflection setting of the ballistic reticle dot with reference to the mil scale of the telescope reticle, and record this measurement as a plus mil value (R+). Utilizing this procedure measure and record the deflection settings of the ballistic reticle for null settings of .33221 and .66442. Record these deflection settings as plus mil values (R+). All rotations of R7 shall be continuous in one direction, without overtravel. Upon completion of the reticle deflection measurement for the null setting of .66442 set a null setting of .83052 on the ratio

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standard and rotate R7 until a null is produced on the scope. Then set a null setting of .66442 on the ratio standard and rotate R7 in the opposite direction (minus) until a null is produced on the oscilloscope. Utilizing this procedure measure and record the reticle deflection settings for null settings of .33221 and .16610. Record these deflection settings as minus mil values (R-). Again using the above procedures obtain and record 2 additional bi-directional (R+ and R-) mil deflection settings for null settings of .16610, .33221, and .66442 on the right side of electrical zero. Upon completion of the measuring and recording of the 3 sets of bi-directional horizontal reticle line of sight readings for the null settings on the right side of electrical zero, return the ballistic reticle to electrical zero by setting the ratio standard to .00000. Position the test circuit phase switch for movement of the ballistic reticle to the left of electrical zero. Measure and record 3 sets of bi-directional (L+ and L-) mil deflection settings for null settings of .16610, .33221, and .66442 on the left side of electrical zero by utilizing the above procedures for measurements of mil deflection settings on the right side of electrical zero. All mil deflection readings (R+, R-, L+ and L-) for each null setting shall be compared with the computer X and Y mil values to determine conformance to the specified tolerance. The reticle traversing accuracy of the projector shall conform to the requirements of 3.7.10.1

NOTE: The measurements of the reticle position, R and L, may be conducted bi-directional to electrical zero. i.e.: The first set of left bi-directional measurements shall be made in sequence to the first set of right bi-directional measurements.

4.6.7.2.13 Zero line of sight repeatability.- This test may be conducted in conjunction with the test in 4.6.7.2.12 by including 3 sets of bi-directional measurements for a null setting of .00000 on the ratio standard. When this test is conducted in conjunction with 4.6.7.2.12, it shall start with a null setting of .16610 on the ratio standard and the test circuit phase switch positioned for movement to the left of electrical zero. With the reticle in this position, the null setting of the ratio standard shall be changed to .00000 and R7 rotated to its null position. The deflection setting of the ballistic reticle shall be measured and recorded. The phase switch shall be changed for movement to the right. The ratio standard shall then be set for .16610, and R7 shall be nulled and the test in 4.6.7.2.12 continued. Three bi-directional deflection settings measurements at electrical zero (.00000 null setting) shall be made. The reticle line of sight at electrical zero shall conform to the requirements of 3.7.10.2.

4.6.7.2.14 Illumination.- The reticle pattern shall be visually inspected in a darkened area. The reticle shall conform to the requirements of 3.7.11. The rheostat knob shall be rotated from the "OFF" position to its clockwise stop, then rotated back to the "OFF" position. Rotation of the rheostat shall conform to the requirements of 3.7.11.

4.6.8 Boresight knob-torque, infinity sight.- Testing equipment conforming to the requirements of 4.5.2.3.5 shall be utilized for this test. The running torque required to rotate each boresight knob throughout its range of rotation shall not exceed the value specified in 3.8.1 for the temperature indicated.

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4.6.9 Boresight knob disengagement, reticle projector.- Testing equipment conforming to the requirements of 4.5.2.3.4 shall be utilized for this test. The force required to disengage each boresight knob from its normal position shall be within the values specified in 3.8.2.1 for the temperature indicated.

4.6.10 Boresight knob torque, reticle projector.- Testing equipment conforming to the requirements of 4.5.2.3.5 shall be utilized for this test. The running torque required to rotate each boresight knob throughout its range of rotation, in its unlocked position, shall be within the values specified in 3.8.2.2 for the temperatures indicated.

4.6.11 Rheostat knob torque.- Test equipment conforming to the requirements of 4.5.2.3.5 shall be utilized for this test. The running torque required to rotate the rheostat knob throughout its range of rotation shall not exceed the value specified in 3.8.2.3 for the temperature indicated.

4.6.12 Continuity.- A continuity check of the projector circuits shall be made using standard measuring equipment to determine conformance with the requirements of 3.7.1.

4.6.13 Dielectric strength.- The projector shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment to determine conformance with the requirements of 3.9.1. The lamps shall be removed for this test.

4.6.14 Insulation resistance.- Upon satisfactory completion of the test in 4.6.13, the projector shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B, of MIL-STD-202. Measurements between insulating points and ground shall be made utilizing standard measuring equipment to determine conformance with the requirements of 3.9.2. The lamps and ground connection shall be removed for this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The projector shall be packaged, packed and marked in accordance with MIL-P-14232/P10516310. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Reticle Image Projector: M14 is used in tank fire control systems in conjunction with a tank periscope and an electronic ballistis computer. Its functions are to project and automatically position laterally, a reticle pattern into the periscope's optical system for control of the main gun; to project a reticle pattern into the unity power viewer of the periscope for control of a machine gun.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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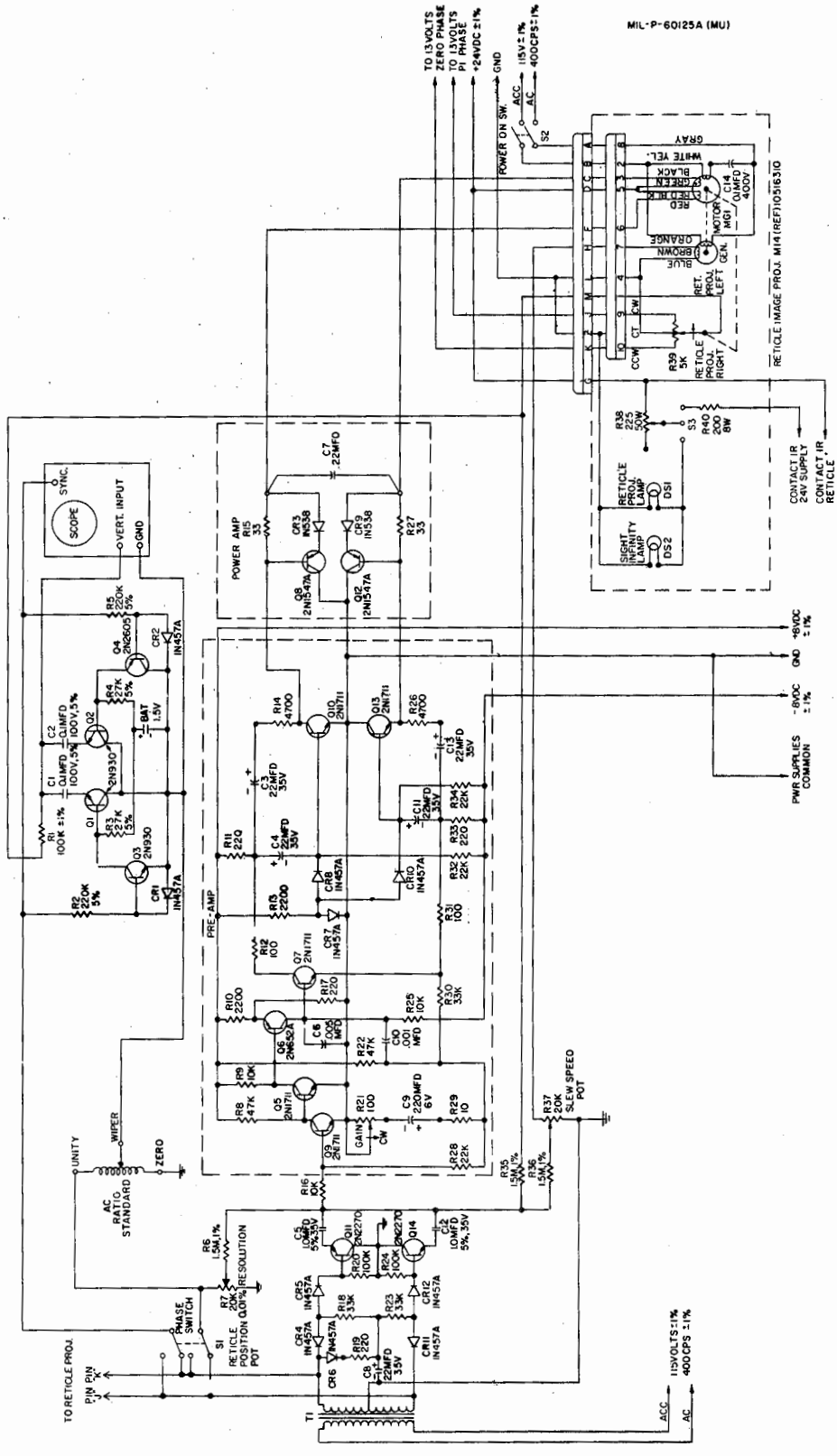


FIGURE 1.

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